Single-Cell RNA-Seq Analysis (PBMC Dataset)

Getting Started with the AnnData Package

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Overview:

This project performs an exploratory analysis of **single-cell RNA sequencing (scRNA-seq)** data using the PBMC dataset. It demonstrates a complete workflow including:

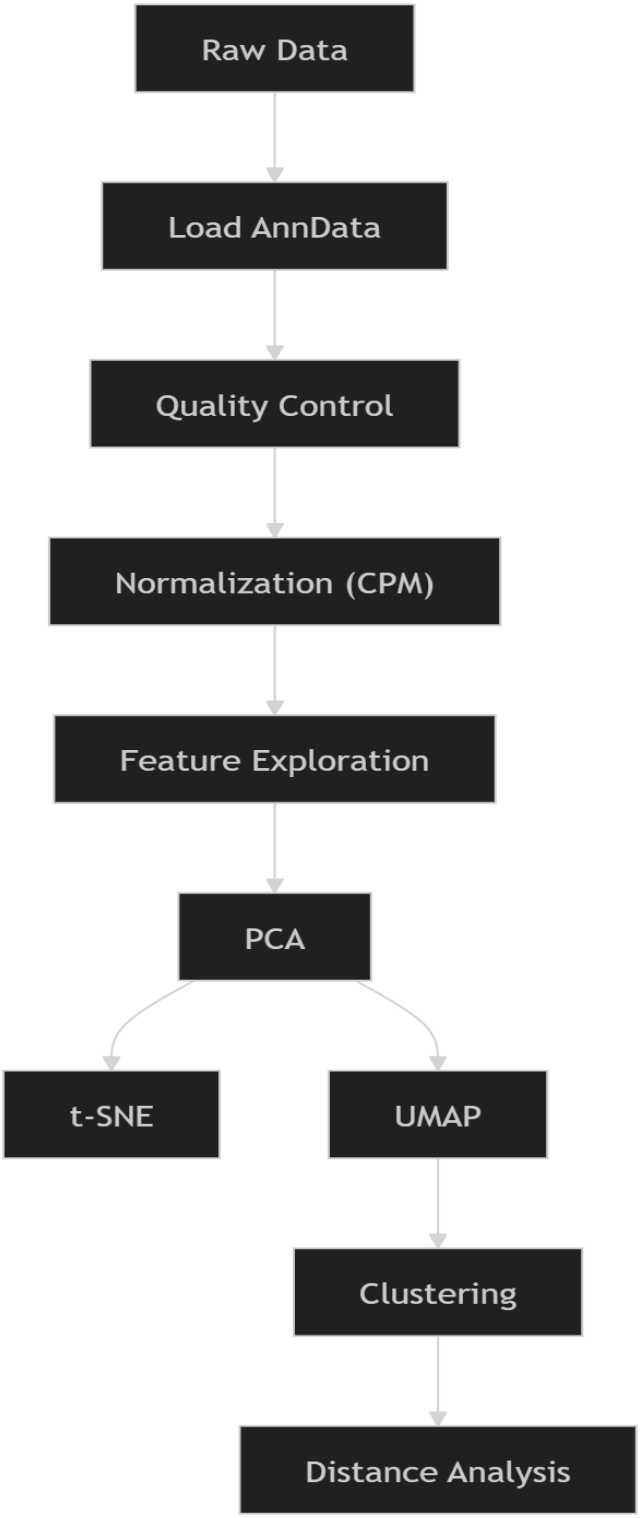
- Data loading and inspection
- Sparse matrix handling
- Normalization (CPM)
- Quality control
- Dimensionality reduction (PCA, t-SNE, UMAP)
- Visualization and clustering

Dataset:

- **Dataset:** PBMC3k
- **Cells:** ~2700
- **Genes:** ~1800
- **Format:** AnnData (.h5ad)

The notebook loads a pre-processed PBMC dataset with 2,638 observations (cells) and 11,505 variables (genes). The active matrix adata.X contains normalized log(1+x)-transformed expression values, and a raw count matrix is preserved in adata.layers["raw"]. The object also stores cell-level metadata in adata.obs, gene-level metadata in adata.var, embedding coordinates in adata.obsm, pairwise distances in adata.obsp, and analysis settings/results in adata.uns.

Workflow:



Results & Visualizations:

1. Gene Expression (Matrix Plot)

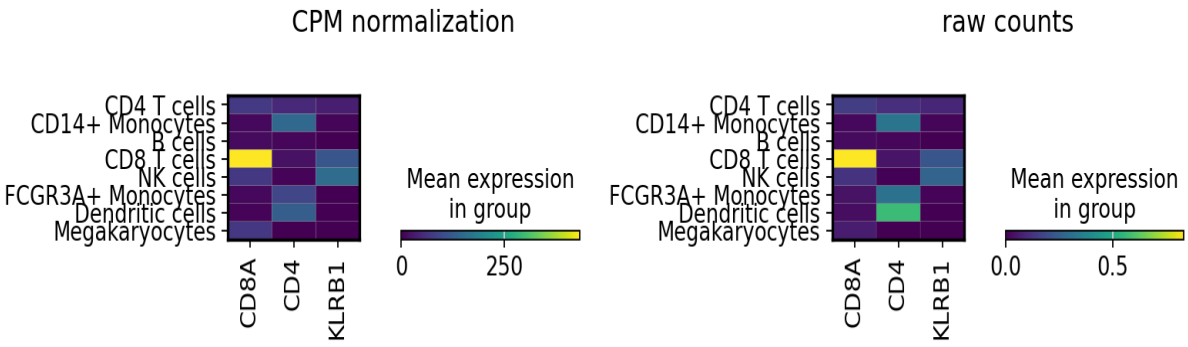


Figure 1. Matrix plot comparing CPM-normalized values with raw counts

Interpretation. This figure compares the average expression of three marker genes (CD8A, CD4, and KLRB1) across louvain-derived cell types using two data representations: counts-per-million normalization and raw counts. The normalized panel makes differences in relative expression more interpretable across groups, while the raw-count panel shows the same patterns on a lower, count-based scale. Together, the figure demonstrates why multiple layers are useful in AnnData: one layer can preserve original counts, while another can support downstream visualization and comparison.

2. Dimensionality Reduction (PCA, t-SNE, UMAP)

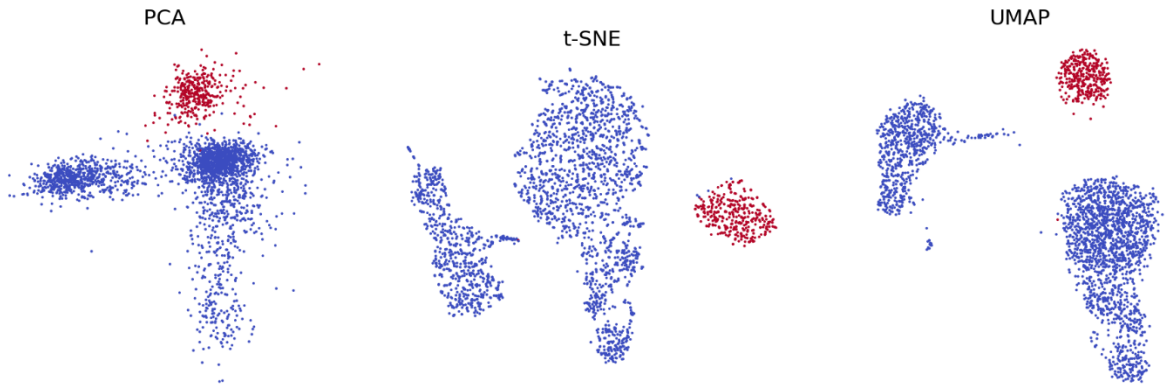


Figure 2. PCA, t-SNE, and UMAP embeddings highlighting B cells

Interpretation. This panel shows three complementary low-dimensional views of the same data. In each embedding, B cells are highlighted in red against the remaining cells in blue. PCA preserves broad linear structure, while t-SNE and UMAP emphasize neighborhood relationships and cluster separation more clearly. The consistent localization of B cells across the three embeddings supports the idea that B cells form a

transcriptionally coherent population distinct from most other immune cell groups in the dataset.

Insights:

- **PCA** → captures variance
- **t-SNE** → highlights local clusters
- **UMAP** → best separation of cell populations
- Clear distinction of **B cells vs others**

3. Distance Matrix Heatmap:

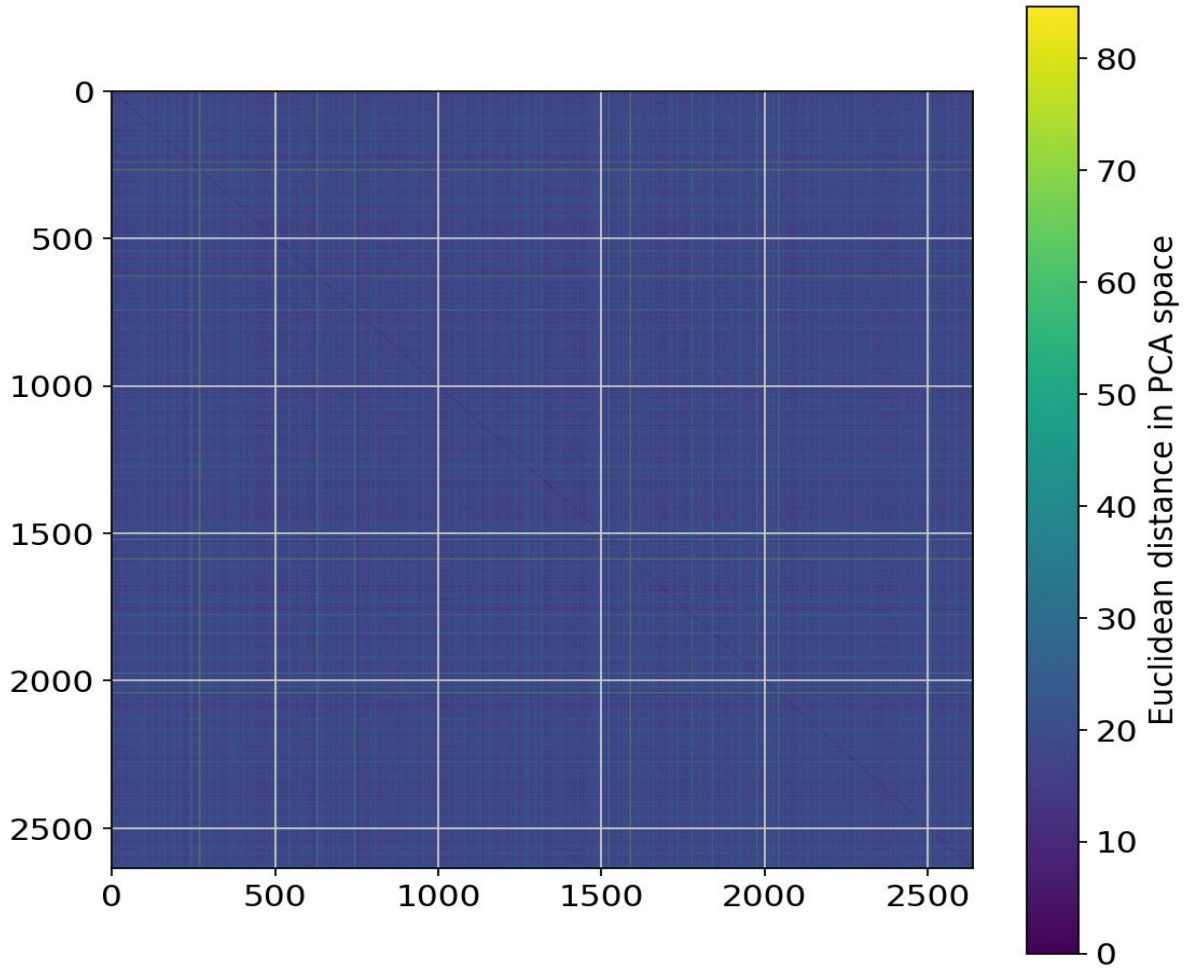


Figure 3. Full pairwise cell-cell distance matrix in PCA space

Interpretation. The first distance heatmap shows Euclidean distances between all pairs of cells in PCA space using the original order of observations. Because the cells are not arranged by type, the matrix appears relatively unstructured, even though subtle vertical and horizontal bands suggest that some groups are more similar internally than others. This is a useful diagnostic plot, but biological structure is difficult to interpret without reordering.

4. Reordered Distance Matrix

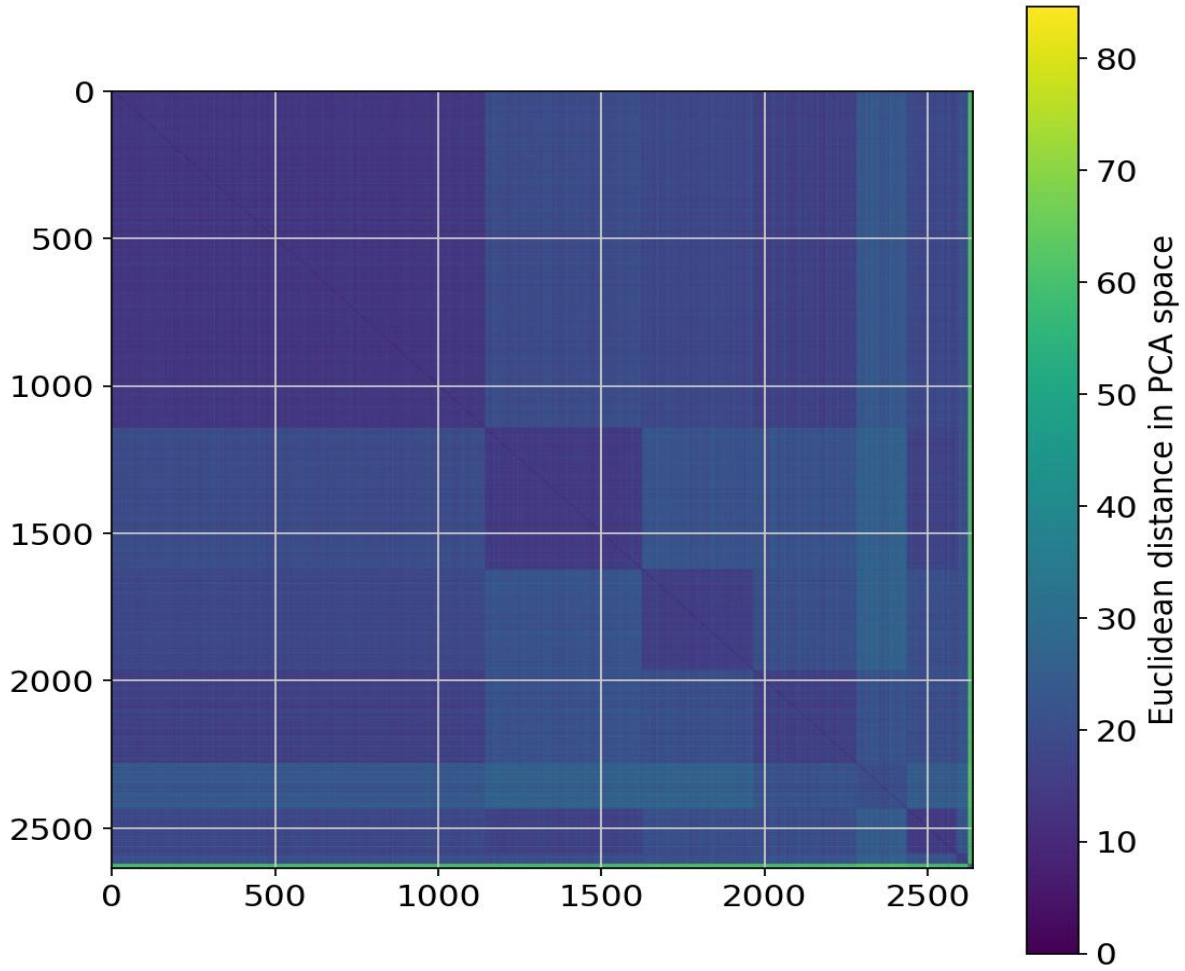


Figure 4. Reordered pairwise distance matrix by louvain cell type

Interpretation. After reordering cells by louvain cell type, clear block-like patterns emerge in the distance matrix. These blocks indicate that cells within the same annotation group are more similar to one another than to cells in other groups. The appearance of this structure confirms that the clustering and annotation metadata capture biologically meaningful separation among PBMC populations.

Metadata and Annotation Review:

How metadata are organized inside AnnData.

- Cell annotations in obs include n_genes, percent_mito, n_counts, and louvain_cell_types.
- A new boolean annotation, is_low_quality, was created by flagging cells with percent_mito > 0.03.
- Gene annotations in var include gene_names, n_cells, and gene_ids.
- obs_names use cell barcodes as the observation index, while var_names initially use gene names and can be switched to gene IDs if desired.

Main Findings:

- The PBMC dataset is sparse but information-rich, making sparse storage efficient.
- Multiple layers are valuable because raw counts and transformed values serve different purposes.
- Cell-type annotations align well with embedding-based structure, especially for B cells.
- Pairwise distance matrices become much more interpretable when cells are reordered by annotation.
- AnnData provides a unified framework for expression values, metadata, embeddings, pairwise relationships, and analysis settings.

Conclusion:

This report provides a clear introductory demonstration of how AnnData structures single-cell RNA-seq data and how Scanpy can be used to inspect, annotate, subset, and visualize that data. The figures show that the PBMC dataset contains well-separated immune populations, that metadata-driven grouping is consistent with geometric structure in low-dimensional space, and that the AnnData object is designed to keep every component of the analysis aligned. It is a strong foundation for moving from data inspection to downstream single-cell analysis.

Google Collab Link:

https://colab.research.google.com/drive/1eePv1aHYlB34Ujy14WduZsvF2Z0xgwQt?usp=sharing